Sahaj Bhadja

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Summary

Frontend-focused Software Engineer with 2+ years building **React** and **TypeScript** single-page apps with data-heavy dashboards, filtering, and forms on REST APIs, tested with **Jest** and Selenium, plus Node.js and Python backend experience.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Web Technologies, Database Systems, Analysis of Algorithms, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Object-Oriented Programming, Operating Systems, Data Retrieval

Experience

USC Dornsife **September 2025 – Present**
Software Engineer *Los Angeles, CA*

- Cut **dashboard latency** from 1.2s to 350ms for 500+ users by restructuring the queries, indexes, and access patterns behind a multi-tenant platform's dashboard APIs on Azure SQL.
- Built FastAPI services and integrations with validation, error handling, and reconciliation checks, shipping CI/CD releases that kept 5K+ monthly jobs and 1K+ AI jobs at 99%+ production **reliability**.

Grad8 **June 2025 – August 2025**
Software Engineer Intern *Los Angeles, CA*

- Built a **React** single-page application with upload, grading history, and question management views on Flask REST APIs, handling async loading and error states for 150+ instructors and 10K+ daily requests.
- Designed the frontend-to-backend flow for file uploads, rubric generation, grading, and history across PDF, DOCX, and ZIP inputs, cutting feedback turnaround 60% and shipping through GitHub Actions **CI/CD**.

Aasma Technology Solutions **August 2023 – July 2024**
Software Developer Engineer – SecureSync *India*

- Wrote 20+ unit and integration **test** suites with PyTest and Mockito in nightly Jenkins pipelines across Windows, Linux, and macOS, generating coverage reports and failure logs for troubleshooting.
- Optimized **API** responses for low-bandwidth clients with pagination, page and limit parameters, and field filtering so clients fetched only needed data, improving response times 30% in Agile sprints.

Aasma Technology Solutions **May 2023 – August 2023**
Full Stack Developer Intern – VibrantChem *India*

- Built **React** and TypeScript CRM interfaces and dashboards for customer profiles, resource allocation, billing, and leads on Node.js REST APIs, with real-time WebSocket updates across 2M+ synchronized records.
- Automated Jest, Mocha, and Selenium unit, integration, and end-to-end **frontend tests** in Jenkins CI/CD, reaching 95%+ build validation, and used Google Analytics load times and drop-off points to improve usability.

LST Capitals **November 2022 – January 2023**
Web Developer Intern *India*

- Designed end-to-end UI/UX flows in Figma with the client, reducing design errors 25%, and supported **React TypeScript** web and React Native clients with Flask APIs and Firebase real-time status updates.
- Built booking, pricing, and coupon APIs with transactional locking and idempotent payment webhooks, sustaining sub-200ms latency across 50K+ booking records and feeding hourly Databricks analytics into internal **dashboards**.

Skills

Frontend: React, TypeScript, JavaScript, Next.js, HTML, CSS, TailwindCSS, Material UI, Shadcn, Angular
UI Engineering: Single-Page Applications, Reusable Components, Async Data Fetching, Forms and Validation, Data Tables and Filtering, Data Visualization, Responsive Design
Testing: Jest, Mocha, Selenium, PyTest, Unit Testing, Integration Testing, End-to-End Testing, Code Reviews
Backend & APIs: Node.js, Express.js, REST APIs, WebSockets, FastAPI, Flask, Python, SQL
Tools & Observability: Git, GitHub Actions, CI/CD, Jenkins, Docker, Sentry, Kibana, Google Analytics

Projects

FinNexus Live | *Next.js, TypeScript, TailwindCSS, WebSockets, MongoDB*

- Built a **TypeScript** and Next.js fintech SPA with an advanced transaction filtering UI, Plaid bank linking, real-time WebSocket updates, and TailwindCSS components for 50K+ users across 1M+ transactions.

HMIS – Hospital Management Information System Live | *TypeScript, React, TailwindCSS, Twilio, Sentry*

- Won Smart India Hackathon with a TypeScript and **React** app featuring doctor and hospital dashboards, appointment booking and rescheduling forms, and responsive TailwindCSS UI, monitored with Sentry across 100+ daily appointments.

CPU-Scheduling Simulator Live | *React.js, JavaScript, EJ-2, C++*

- Built an interactive React simulator with EJ-2 Gantt chart **data visualization**, custom process input forms, and live waiting and turnaround metrics for six scheduling algorithms across 1K+ process combinations.